SIMATS ENGINEERING

CO1-ASSESSMENT TOOL 2-Pseudocode Writing Task

COURSE NAME: R PROGRAMMING **COURSE CODE: ITA04**

COURSE OUTCOME COVERED

CO: Apply R programming fundamentals including data structures, vectors, matrices, arrays, and class types to perform mathematical operations and manage data effectively.
(Bloom Level -BL2- BL4)

Assessment Tool Weightage: 20%

Code of Conduct:

*I **YUGITHA B (192424102)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

1. Variables and Data Types Analysis

Pseudocode

```
BEGIN

   DECLARE num1 AS NUMERIC
   DECLARE num2 AS NUMERIC
   DECLARE name AS CHARACTER
   DECLARE isPassed AS LOGICAL

   SET num1 = 20
   SET num2 = 10
   SET name = "Yugitha"
   SET isPassed = TRUE

   // Arithmetic operations
   SET addition = num1 + num2
   SET subtraction = num1 - num2
   SET multiplication = num1 * num2
   SET division = num1 / num2

   // Logical operation
   SET result = (num1 > num2) AND isPassed

   DISPLAY "Name:", name
   DISPLAY "Addition:", addition
   DISPLAY "Subtraction:", subtraction
   DISPLAY "Multiplication:", multiplication
   DISPLAY "Division:", division
   DISPLAY "Logical Result:", result

END
```

2. Data Structures Implementation

Pseudocode

```
BEGIN

   // Create and initialize a vector
   CREATE vector = [10, 20, 30, 40, 50]

   // Indexing
   DISPLAY vector[1]
```

```
// Updating
SET vector[2] = 25

// Aggregation
SET total = SUM(vector)
DISPLAY total


// Create and initialize a list
CREATE list = [
   Name = "Yugitha",
   Age = 20,
   Marks = 85
]

// Indexing
DISPLAY list[Name]

// Updating
SET list[Marks] = 90

DISPLAY list


// Create and initialize a matrix
CREATE matrix =
   [10 20
    30 40]

// Indexing
DISPLAY matrix[1,2]

// Updating
SET matrix[2,1] = 35

// Aggregation
SET matrixTotal = SUM(matrix)

DISPLAY matrix
DISPLAY matrixTotal

// Comparison
DISPLAY "Vector: Stores similar type of values"
DISPLAY "List: Stores different types of values"
```

```
    DISPLAY "Matrix: Stores data in rows and columns"

END
```

3. Matrix and Array Operations

Pseudocode

```
BEGIN

    // Create Matrix A
    CREATE Matrix_A =
        [1 2
         3 4]

    // Create Matrix B
    CREATE Matrix_B =
        [5 6
         7 8]

    // Matrix Addition
    SET addition = Matrix_A + Matrix_B

    // Matrix Multiplication
    SET multiplication = Matrix_A %*% Matrix_B

    // Element-wise multiplication
    SET elementwise = Matrix_A * Matrix_B

    DISPLAY "Matrix A:"
    DISPLAY Matrix_A

    DISPLAY "Matrix B:"
    DISPLAY Matrix_B

    DISPLAY "Matrix Addition:"
    DISPLAY addition

    DISPLAY "Matrix Multiplication:"
    DISPLAY multiplication

    DISPLAY "Element-wise Multiplication:"
    DISPLAY elementwise
```

```
    // Create 3D Array
    CREATE Array_3D with dimensions 2 × 2 × 2

    SET Array_3D values =
        [1, 2, 3, 4, 5, 6, 7, 8]

    DISPLAY "3D Array:"
    DISPLAY Array_3D

    // Access an element
    DISPLAY Array_3D[1,1,1]

    DISPLAY "Analysis:"
    DISPLAY "Matrix is used for two-dimensional data."
    DISPLAY "3D Array is used for data with three dimensions."

END
```

4. Data Frame Construction and Manipulation

Pseudocode

```
BEGIN

    // Create data frame
    CREATE data frame StudentData with:

        ID = [101, 102, 103, 104, 105]

        Name = ["Arun", "Bala", "Divya", "Kavi", "Ravi"]

        Age = [20, 21, 20, 19, 21]

        Marks = [80, 75, 90, 65, 85]

    DISPLAY StudentData


    // Add a new column
    ADD column Department =
        ["AI&DS", "CSE", "AI&DS", "ECE", "CSE"]

    DISPLAY StudentData
```

```
    // Modify a column
    SET Marks =
        [85, 78, 92, 70, 88]

    DISPLAY StudentData


    // Delete a column
    DELETE column Age

    DISPLAY StudentData


    // Summary operations
    SET average = MEAN(Marks)
    SET maximum = MAX(Marks)
    SET minimum = MIN(Marks)

    DISPLAY "Average Marks:", average
    DISPLAY "Maximum Marks:", maximum
    DISPLAY "Minimum Marks:", minimum

END
```

5. Functions and Logical Operations

Pseudocode

```
BEGIN

    // Define a user-defined function
    FUNCTION CalculateTotal(a, b, c)

        SET total = a + b + c

        RETURN total

    END FUNCTION


    // Input marks
    SET maths = 80
    SET science = 75
    SET computer = 90
```

```
    // Call function
    SET totalMarks =
        CalculateTotal(maths, science, computer)

    DISPLAY "Total Marks:", totalMarks


    // Logical vector
    SET marks = [85, 45, 72, 90, 38, 65]

    // Students who scored above 50
    SET passed = marks >= 50

    DISPLAY "Pass/Fail:"
    DISPLAY passed

    // Display only passed marks
    SET passedMarks = marks[marks >= 50]

    DISPLAY "Passed Marks:"
    DISPLAY passedMarks


    // Students who scored above 75
    SET highMarks = marks[marks > 75]

    DISPLAY "Marks above 75:"
    DISPLAY highMarks

END
```